CoRoT-2b

R-1131

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_CoRoT-2_190526` · created 2026-07-30T23:01:17+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458629.8844 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.218 +/- 0.066
Transit depth (Rp/Rs)^2	4.76 +/- 2.88 [%]
Orbital Inclination (inc)	89.26 +/- 2.03
Airmass coefficient 1 (a1)	0.81 +/- 0.58
Airmass coefficient 2 (a2)	0.12 +/- 0.39
Scatter in the residuals of the lightcurve fit is	69.03 %
Best Comparison Star	#2 - [199, 23]
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.0993 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.218 ± 0.066 vs 0.1667 (deviation 0.78σ)
Duration fit vs published	0.0993 d vs 0.095 d (deviation 0.65σ)
Mid-time O–C	-6.5 min
Depth SNR	3.3
Residual scatter	69.03 %

Light curve

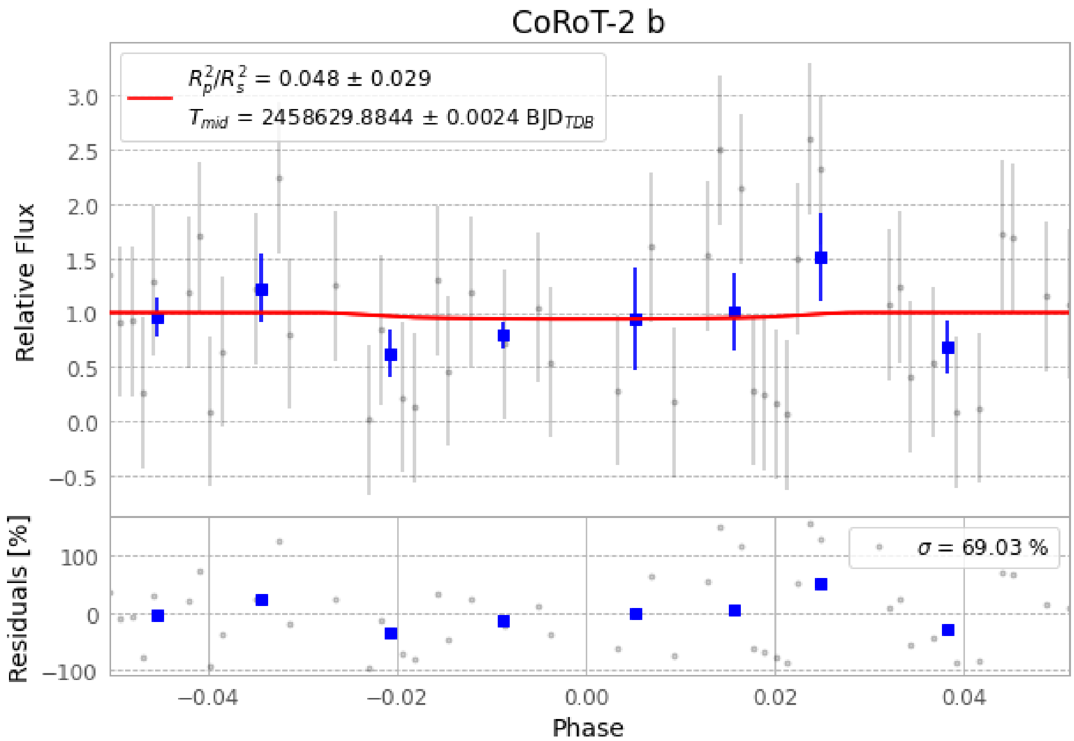

Figure 1 — FinalLightCurve_CoRoT-2 b_2019-05-26.png (SHA-256 ddeea30d7b7e52eb…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [125, 273], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3365996641edf1dc…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

86 FITS frames (56 MB), `CoRoT-2190526070012.FITS` → `CoRoT-2190526111512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-190526.log (4cb2efb34e50…), FinalLightCurve_CoRoT-2 b_2019-05-26.png (ddeea30d7b7e…), FinalLightCurve_CoRoT-2 b_2019-05-26.csv (6321cc342c27…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)